3 **(a)** **(i)** One of the assumptions of the kinetic theory of gases is that the motion of gas molecules is random. Therefore explain why the average momentum of the gas molecules in a container is zero.

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..... [2]

(ii) According to the kinetic theory of gases, explain how a gas exerts a pressure on the walls of a container.

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..... [3]

(b) A student designed an air-filled "load-lifting" system where the air could be uniformly heated by suitably located electrical heaters in the enclosure of the system. All sides of the enclosure have good thermal insulation and the top surface is movable vertically.

The volume of air inside the enclosure at tempature of 300 K and pressure of 101.0 kPa is 2.000 m^3.

When a load of mass 500 kg is placed on the top of the movable surface of the system, an average depression of 5.0 cm of the top surface is observed and the volume of air in the enclosure is reduced to 1.947 m^3.

(i) State the first law of thermodynamics.

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..... [2]

(ii) The air may be considered as an ideal gas. Determine

1. the change in internal energy of the air in the system after the load is placed on top of the movable surface.

change in internal energy = …….. J [3]

2. the new pressure of air in the system.

new pressure = …….. kPa [3]